

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:**Part i**

Interest rate	12%				
starting increase	4%				
	0.50%				
Recovery	40.00%				
Year		Default per 100	Survival		Present value
1	4%	4.0000	96.0000	0.8869	0.8514
2	4.50%	4.3200	91.6800	0.7866	0.7212
3	0.00%	-	91.6800	0.6977	0.6396
4	5.50%	5.0424	86.6400	0.6188	0.5361
					2.7484

		Periodic payment		2.7484S	
Principal payment		default take place mid of year			
Year	Default per 100	Recovery	Pay off	Discount	Present value
1	4	40.00%	2.4000	0.9418	0.0226
2	4.32	40.00%	2.5920	0.8353	0.0217
3	0	40.00%	-	0.7408	-
4	5.0424	40.00%	3.0254	0.6570	0.0199
					0.0641
		Total principal payment		0.0641	

Accrual adjustment per 1 unit of spread

Year	Survival	Accrual	Discount	
1	96.0000	2.0000	0.9418	0.0188
2	91.6800	2.1600	0.8353	0.0180
3	91.6800	-	0.7408	-
4	86.6400	2.5212	0.6570	0.0166
			Sum	0.0534

Accrual payment 0.0534S

Spread 0.02288 0.0641/(2.7484+0.0534)
228.7815

Part ii

Interest rate 12%
 starting 4%
 increase 0.50%
 Recovery 40.00%

Year		Default per 100	Survival		Present value
1	4%	4.0000	96.0000	0.8869	0.8514
2	4.50%	4.3200	91.6800	0.7866	0.7212
3	5.00%	4.5840	87.1000	0.6977	0.6076
4	5.50%	4.7903	82.3100	0.6188	0.5093
					2.6896

		Periodic payment		2.6896S	
Principal payment	default take place mid of year				
Year	default per 10	Recovery	Pay off	Discount	Present value
1	4.0000	40.00%	2.4000	0.9418	0.0226
2	4.3200	40.00%	2.5920	0.8353	0.0217
3	4.5840	40.00%	2.7504	0.7408	0.0204
4	4.7903	40.00%	2.8742	0.6570	0.0189
					0.0835
Total principal payment			0.0835		

Accrual adjustment per 1 unit
 of spread

Year	Survival	Accrual	Discount	
1	96.0000	2.0000	0.9418	0.0188
2	91.6800	2.1600	0.8353	0.0180
3	87.1000	2.2920	0.7408	0.0170
4	82.3100	2.3951	0.6570	0.0157
		Sum		0.0696

Accrual payment 0.0696S

Spread $0.0303 = (0.0835 / (0.0693 + 2.6896))$
 302.6239

[6]
 [14 Marks]

Solution 2:

i)

a) Reasons for choosing a full yield curve model:

The rationale for a term structure yield curve model is to be able to price simultaneously options spread across the entire range of maturities in a yield curve.

Exotic interest rate swaps and options include spread options, Bermudan swaptions and path dependent options. These depend not just on the evolution of forward rates along the curve, but on the correlation between changes.

[3]

b) Desirable features:

- Model is arbitrage free, so produces prices consistent with the market
- Flexible enough to cope properly with the required range of derivative contracts
- Easy to specify and calculate
- The current swap (or bond) curve should be reproduced by the model
- Easy to calibrate
- Enough degrees of freedom (parameters) to make the model flexible to cope with any yield curve shape, but not overly flexible so there is instability between parameters from one day to the next
- Volatility of rates of different maturity should be different, with shorter rates usually seeing higher volatility
- Imperfect correlation between forward rates
- Negative interest rates should not normally be allowed
- Mean reversion of interest rates
- Reasonable dispersion of rates over time

[3]

c) No – arbitrage models:

No-arbitrage, or “arbitrage free”, models are a class of models that allow recovery of market prices of one set of securities given prices of another set. This creates a world of “relative” prices.

In a non-“arbitrage free” model, securities could be priced using the model and then traded at a different price in the real world, leading to persistent profit – something not tenable in the financial markets.

No-arbitrage is very important in yield curve models, since most complex structures are limiting cases of simpler structures (such as swaps, caps, floors) and ideally the model should recover the prices of the latter exactly.

Also, hedging is done using the simpler structures, so the accounting process will not be distorted by imaginary gains and losses.

[3]

- ii) A trinomial tree is a discrete-time representation of the continuous rate process, with the stochastic process defined by branches which can have three states: up, down and mid-way between.

If the probabilities of going up, down and middle are p_u , p_d and p_m respectively, then $(p_u + p_d + p_m) = 1$ and all values are strictly positive.

The time horizon is split into constant intervals of time Δt , so the rates are compounded Δt period rates.

Further, accuracy is improved with smaller time-steps, up to a point.

The tree for the one-factor model is built in two stages.

Stage 1:

First, set $b(t)$ to zero. Then,

$$dr^* = -ar^*dt + \sigma dz$$

which gives a rate process r^* that is initially zero, and whose evolution is governed by a constant mean reversion towards zero.

The value of σ is determined from the caplet volatilities.

There are three unknowns: p_u , p_d and Δr^* (since $p_m = 1 - p_u - p_d$).

It is convenient to set $\Delta r^* = \sigma\sqrt{3\Delta t}$, since this has been found to give the best numerical efficiency.

The other two equations come from the expectation and variance of Δr^* over the interval Δt , in which the tree must match the process.

There are three different forms of the mean and variance equations for the tree: when all the branches point upwards or horizontal, when all point downwards or horizontal, and when they straddle the line of unchanged rates.

There will be different probabilities for each situation.

Since $E(\Delta r^*)^2$ contains the term $\sigma\Delta t$, the equations account for the volatility component.

Stage 2:

Now add back in the term structure, to move back from r^* to r .

Let $\alpha(t) = r(t) - r^*(t)$. Then:

$$d\alpha = a[b(t) - \alpha(t)]dt \quad (1)$$

which is a deterministic mean-reverting process (i.e., varies over time but not stochastically), which can easily be calibrated to the term structure.

The best method of solving the differential equation in (1) is by forward induction along the Δt scale using Green's functions. A Green's function $Q_{i,j}$ is the present value of 1 unit of cash payable at node (i, j) on the tree.

Now, $Q_{0,0} = 1$ and $\alpha_0 (= r)$ is set in terms of the price of a zero-coupon bond maturing at Δt .

Then $Q_{1,1}$, $Q_{1,0}$ and $Q_{1,-1}$ are set in terms of the tree probabilities p_u , p_d and p_m , $Q_{0,0}$ and α_0 .

One can then easily find the value of α_1 based on the price of a zero-coupon bond maturing at $2\Delta t$ and these Q 's.

One can then move up the time horizon.

[5]

iii) Does HW fulfil the desirable features?

The model is a suitable no-arbitrage full yield curve model as

- It reproduces vanilla bond and swap prices exactly and prices options on these
- The model incorporates mean reversion and time-dependent parameters, so is flexible, behaves well and has little chance of producing negative rates
- It is relatively easy to specify mathematically, though some care is required.

However, a few issues that need to be mentioned are

- It only has one driving factor, so all forward rate moves have to be completely correlated and hence it cannot re-create complex yield curve changes
- The underlying distribution of the interest rate is normal, not log-normal, so it can lead to negative rates [theoretically, as in practice the probabilities of negative rates are negligible]
- The model may be hard to calibrate to cap prices which are priced in the market with the Black's log-normal model.

Also,

- Differences arise as interest rates move away from the current level. These effects, if not disentangled, will lead HW to have an incorrect balance between its volatility and mean-reversion parameters.
- Using constant values for α and σ limits the shapes of yield curve that can be modelled. Time-dependent mean reversion and volatility parameters can fit any current yield curve and forward volatility "hump" shape although they can imply an implausible evolution of the term structure of volatility.

[5]

[19 Marks]

Solution 3:

i)

a) Cross hedging is the practice of hedging exposure to the price of one asset using futures with a different underlying asset. [1]

b) The hedge ratio is the ratio of the position taken in futures to the size of the exposure. [1]

- c) The minimum variance hedge ratio is the hedge ratio that minimizes the variance of the hedger's net position. [1]
- d) Tailing the hedge is a small adjustment to make to the hedge to allow for how futures (with daily settlements and $\Delta \neq 1$ rather than forward deals with $\Delta = 1$) are being used as hedging instruments. [1]

ii) Minimum variance hedge ratio h , is given by

$$h = \frac{\rho\sigma_S}{\sigma_F}$$

where, $h = 0.75 * (4.2/2.5) = 1.26$.

Therefore, the number of futures contracts to buy, before tailing the hedge, is

$$1.26 * (100000 / 1000) = 126.$$

Thus, the hedge can be tailed by

$$126 * (101.20/106.30) = 119.95 \text{ or approximately } 120 \text{ ATF futures contracts.}$$

[2]

iii) Let r be the riskfree rate and u be the rate of storage costs. Then, the t – year futures price is given by

$$F_t \leq S e^{(r+u)t}.$$

If this is not satisfied, investors could make arbitrage profits by borrowing cash, buying oil and shorting oil futures.

However, in certain circumstances where physical storage capacity (or indeed borrowing capability) is limited, prices can remain distorted despite this arbitrage occurring.

One cannot form a no arbitrage argument from the opposite side of the arbitrage and prove

$$F_t = S e^{(r+u)t}$$

This is because,

1. There are not enough investors holding the commodity as an investment (and ready to take arbitrage profits by selling the commodity and buying futures)
2. There are benefits in holding the commodity physically, meaning that people holding the commodity physically benefit from the “convenience yield”

Thus, futures prices can only give us a lower bound on commodity storage costs of

$$u \geq \frac{1}{t} \ln \left(\frac{F_t}{S} \right) - r.$$

[4]

[10 Marks]

Solution 4:

i)

Total return swap is a contract where 2 firms decide the swap overall return from assets. This is used to gain exposure to an asset without buying or selling transaction. Value of TRS at start of the transaction is zero .

[2]

ii)

Amortising bond with 0 redemption value would have total N payments, with the first payment at $t = 1$ and the last at $t = N$. We can express the present value of bond as the sum of the present values of each individual annuity payment, as follows:

$$PV = \frac{A}{(1+r)} + \frac{A}{(1+r)^2} + \frac{A}{(1+r)^3} + \dots + \frac{A}{(1+r)^{N-1}} + \frac{A}{(1+r)^N}$$

where

A = annual payment amount

r = the interest rate per period corresponding to the frequency of payments (for example, annual, quarterly, or monthly)

N = the number of annuity payments

Because the payment (A) is a constant in this equation, it can be factored out as a common term. Thus the sum of the interest factors has a shortcut expression:

We use

$$PV = A \left[\frac{1 - \frac{1}{(1+r)^N}}{r} \right]$$

[2]

iii)

2 for estimating the payout we first have to calculate the pay out at $t=0$ and then estimate principle at $t=3$ then use that principle and new rate to calculate annual pay out for asset B

This is shown in below calculation (Values in '000)

		Asset 1	Asset 2
	PV	1,00,000	1,00,000
	r	5%	3%
	term	30	30
A	$(1+r)^n$	4.32	2.43
B	$1/A$	0.23	0.41
C	$1-B$	0.77	0.59
D	C/r	15.37	19.60
	Coupon= PV/D	6505.1	5101.9

Asset 2 new Principle at year 3				
	Amount	Coupon	Interest	Amount remaining
t=0	100000	5101.9	3000	97898.1
t=1	97898.1	5101.9	2937	95733.1
t=2	95733.1	5101.9	2872	93503.2
t=3	93503.2			

New pay out

		Asset 2 t=3 payout
	PV	93,503
	r	5.25%
	term	27
A	$(1+r)^n$	3.98
B	1/A	0.25
C	1-B	0.75
D	C/r	14.26
	Coupon= PV/D	6555.6

Total pay out

total pay out at t= 4 -50.5

[5]

[9 Marks]

Solution 5:

- i) The expected payoff based on risk-neutral probabilities is discounted at the risk-free rate. Risk-neutral probabilities are used, and discounting is at the risk-free rate. There is no risk premium incorporated into option pricing because of the use of arbitrage [2]
- ii) The up and down factors express how high and how low the underlying can go. Standard deviation does not appear directly in the binomial model, although it is implicit. The ratio of the underlying value to the exercise price expresses the moneyness of the option [2]

- iii) The actual probabilities of the up and down moves are irrelevant to pricing options. The risk-free and exercise price are, of course, highly relevant. [2]

iv)

Probability for the risk neutral binomial tree for the above stock is given by below formula

$$p = \frac{e^{rt} - d}{u - d}$$

This is derived by comparing the expected value of derivative which should be *Expected value* = $S_0 \times \exp(rt)$

S	100
σ	20%
t	0.25
Sqrt (t)	0.5
u	1.1052
d=1/u (given recombining)	0.9048
K	105
interest rate	5%
$\exp(rt)$	1.0126
p (up)	0.5378
1-p	0.4622
$\exp(-rt)$	0.9876

Time 1			Time 2	
Option value Stock price			Stock price	122.14
	Option Discounted	9.104	Option exercise value	17.14
	Stock price	110.52		
	Option exercise value	5.52		
	100		Stock price	100
			Option exercise value	0
	Stock price	90.48		
	Option exercise value	0		
			Stock price	81.87
			Option exercise value	0

[4]

[10 Marks]

Solution 6:

i)

Forward price is the price which would be paid for an asset after expiration of the contract.

The value of a forward contract at expiration is the spot price of the underlying minus the forward price agreed to in the contract.

$$F_0(T) = (S_0 - \gamma + \theta)(1 + r)^T$$

or

$$F_0(T) = S_0(1 + r)^T - (\gamma - \theta)(1 + r)^T$$

Or for continuous

$$F_0 = S_0 e^{T(r - \gamma + \theta)}$$

Where

$r = \text{interest rate}$

$\gamma = \text{cost of storage}$

$\theta = \text{cash return / dividend}$

[3]

ii) Forward value

$$F(T) = \text{Spot} \times \exp(t(\text{interest} - \text{storage} + \text{collateral}))$$

Forward value	
r	5%
s0	1000
storage	3%
Cash	2%
Forward value	1010.05017

[2]

iii)

Forward rate agreements (FRA) are over-the-counter contracts between parties that determine the rate of interest to be paid on an agreed-upon date in the future. In other words, an FRA is an agreement to exchange an interest rate commitment on a notional amount.

Long position in above FRA means that in 30 days, payment corresponding to the discounted difference between 90-day Libor on that day and the FRA rate on nominal would be exchanged

To do the latter, we would go long a 120-day Mibor time deposit and short a 30-day Mibor time deposit with same nominal

[3]

iv)

Consider 2 portfolio

$$\text{Cash portfolio} = \text{Amount} \times \text{Spot} \times e^{(-0.5 \times R_{us})}$$

$$\text{Forward contract} = A \times F_{(usd/Inr)} \times e^{(-0.5 \times R_{ind})}$$

Both the portfolio have to be equal hence value would be

$$F_{(usd/Inr)} = S e^{(-0.5(R_{us} - R_{ind}))}$$

[3]

v)

INR/USD	75
S	0.0133
r ind	6%
r ud	3%
time	0.5
$F = S \exp(-t * (r_{ind} - r_{us}))$	0.01313
INR/USD	76.1334

[2]

[13 Marks]

Solution 7:

- i) An alternative to decomposing yield curve risk into sensitivity to changes at various maturities (key rate duration) is to decompose the risk into sensitivity to the following three categories of yield curve movements:

Level (Δx_L) – A parallel increase or decrease of interest rates.

Steepness (Δx_S) – Long-term interest rates increase while short-term rates decrease.

Curvature (Δx_C) – Increasing curvature means short- and long-term interest rates increase while intermediate rates do not change.

It has been found that all yield curve movements can be described using a combination of one or more of these movements.

Simple linear model to represent the change in portfolio is where X are the changes in the factor D is weights attached to each component.

[5]

- ii) Linear model can be factor based weighted average of the various interest rate factors

$$\frac{\Delta P}{P} \approx -D_L \Delta x_L - D_S \Delta x_S - D_C \Delta x_C$$

[2]

- iii) The model can be calibrated using statistical technique to break the existing yield curves in above 3 factors. We would have to estimate the expected change in the value of the assets for the above yield curves. This would give us linear regression formula which can be used to calibrate the parameters.

[2]

[9 Marks]

Solution 8:

i)

Effective duration is the sensitivity of a bond's price against the benchmark yield curve.

The effective duration figure is used for hybrid securities, which can be divided into a bond and an option (callable bonds). Embedded bonds increase the uncertainty of cash flows and make it difficult for investors to measure the internal rate of return. This is where the concept of effective duration comes into effect.

When bonds offer an uncertain cash flow, the effective duration is the best way to calculate the volatility of interest rates. The formula is as follows:

$$\text{effective duration} = ED = \frac{BV_{-y} - BV_{+y}}{2 \times BV_0 \times y}$$

$$\text{effective convexity} = EC = \frac{BV_{-y} + BV_{+y} - (2 \times BV_0)}{BV_0 \times y^2}$$

where:

y = change in required yield, in decimal form

BV_{-y} = estimated price if yield decreases by y

BV_{+y} = estimated price if yield increases by y

BV_0

= initial observed bond price

[3]

- ii) Both call and put option reduces the effective duration of the bond

[2]

- iii) A callable bond exhibits positive convexity at high yield levels and negative convexity at low yield levels. Negative convexity means that for a large change in interest rates, the amount of the price appreciation is less than the amount of the price depreciation.

[2]

[7 Marks]

Solution 9:

- i) A leveraged ETF can magnify the index returns using borrowing either directly or through using appropriate derivatives. This can be done by

- Borrowing against the securities and in the portfolio and redeploying the proceeds
- Using options, futures, swaps etc.

[2]

- ii) A leveraged ETF magnifies the daily returns of the index. There is a significant risk of permanent capital loss due to market volatility. Hence, it is not really a suitable product for long term investors. It will be

appropriate for traders, momentum investors and other short-term investors who have a clear view on the short-term direction of the index and capitalise on it. The investor group preferring this product are expected to have sound knowledge of the market and appropriate risk management strategies in place.

[2]

iii) Advantages

- Investors can benefit from the utilisation of leverage without directly taking a position in derivatives.
- No need of arranging for margin funds.
- Being an ETF, the investor can buy and sell the instrument in the market and hence adequate liquidity is available.
- Ability to make amplified returns if the investor view on index direction is correct.
- It is a product which can be accessible to even small investors, who otherwise don't have access to or capability to handle derivatives

Disadvantages

- If the investor view on the market turns out to be wrong, then the amplification offered by leverage can lead to significant capital erosion
- Unlike a normal ETF, the leveraged ETF can have significantly higher expense ratio as there is cost of maintaining the derivative exposure, borrowing costs, etc. which can all cut into the returns offered by the fund.
- The portfolio turnover may also be higher leading to significant attrition costs that can also eat into the returns
- Risk of uneducated investors getting into the products without fully understanding the risks.
- Being a niche product the ETF may not be regularly traded in the exchanges even if it is listed. Then there is an illiquidity risk. Significant bid-ask spreads can also eat into the returns.

[5]

[9 Marks]
